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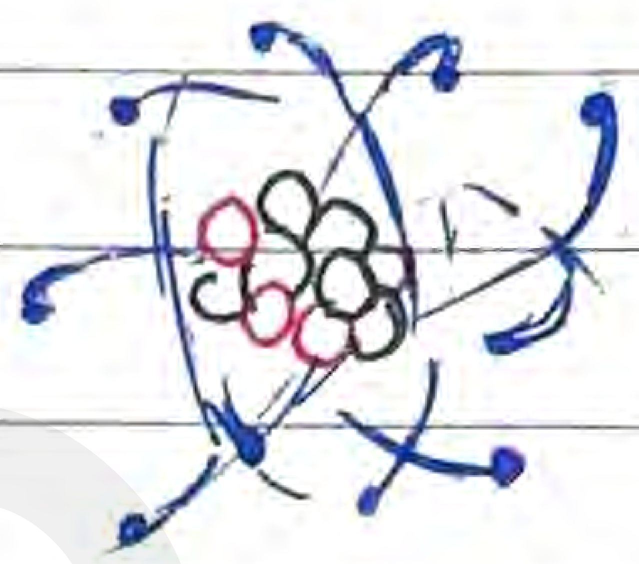
Nuclear Physics



e^- electrons

p^+ protons

n^0 neutrons

$m_{p^+} \approx m_{n^0} \approx 1800\, m_{e^-}$ ← negligible mass

Charges are found in the electron cloud (-) or nucleus (+)

Notation

$${}^{A}_{Z}X$$

Z - Atomic #, # of protons in the nucleus

N - Neutron #, # of neutrons in the nucleus

A - Atomic mass #, # of neucleons in the nucleus

Nucleus consists of neutrons and protons

neutrons $= n = A - Z$

Nucleus (centre of atom)

· Nuclei (multiple nucleus) · Nuclide (distinct type of nucleus)

Isotopes - Same element but of different # of neutrons

$\boxed{\text{Hydrogen...!}}$

${}^{1}_{1}H$ (Hydrogen)	${}^{2}_{1}H$ (Detereum)	${}^{3}_{1}H$ (Tritium)
$1p^+$ $0n^0$	$1p^+$ $1n^0$	$1p^+$ $2n^0$

Nuclear Physics

Evidence for neutrons

- Isotopes – No other way to explain mass difference of two isotopes of the same element
  - ↳ Same element = same # of p^+ = same p^+ mass
  - ↳ As such, must be some other particle as well (n^0)

Summary of Masses

atomic mass unit ↓ (use all s.f. bc of imprecision)

Particle	kg	u	MeV/c^2
Proton	1.6726×10^{-27}	1.007276	938.28
Neutron	1.6750×10^{-27}	1.008665	939.57
Electron	9.109×10^{-31}	5.486×10^{-4}	0.511
^1_1H atom	1.6735×10^{-27}	1.007825	938.78

MeV/c^2 is from . . .

$eV = \text{Energy} = mc^2$

Thus $m = \frac{eV}{c^2}$ with prefix $M = 10^6$

atomic mass unit is also $g \cdot mol^{-1}$ (periodic table given

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Binding Energy

In the macro world, combining two objects together yields a mass equal to the sum of the two smaller objects

↳ But in the nuclear world, every tiny bit of mass matters

↳ when you combine two objects, they release energy (making bonds = exothermic) ⇒ Nucleus

↳ To release energy, mass is converted (the mass converted into energy is known as mass defect)

that's why ${}_1H$ has no binding energy

$E = mc^2$ ← conversion mass to energy

In the nuclear world, every energy released matters since $m = \frac{E}{c^2}$ is of similar magnitude to other masses. → in the macro world, masses are already big so Δm negligible

$\boxed{\text{Thus...}}$ separate = together + energy released

* Protons + Neutrons > mass Nucleus

Mass Defect = Δmass = (Protons + Neutrons) − Nucleus

↳ Energy Released = $\Delta m \cdot c^2$

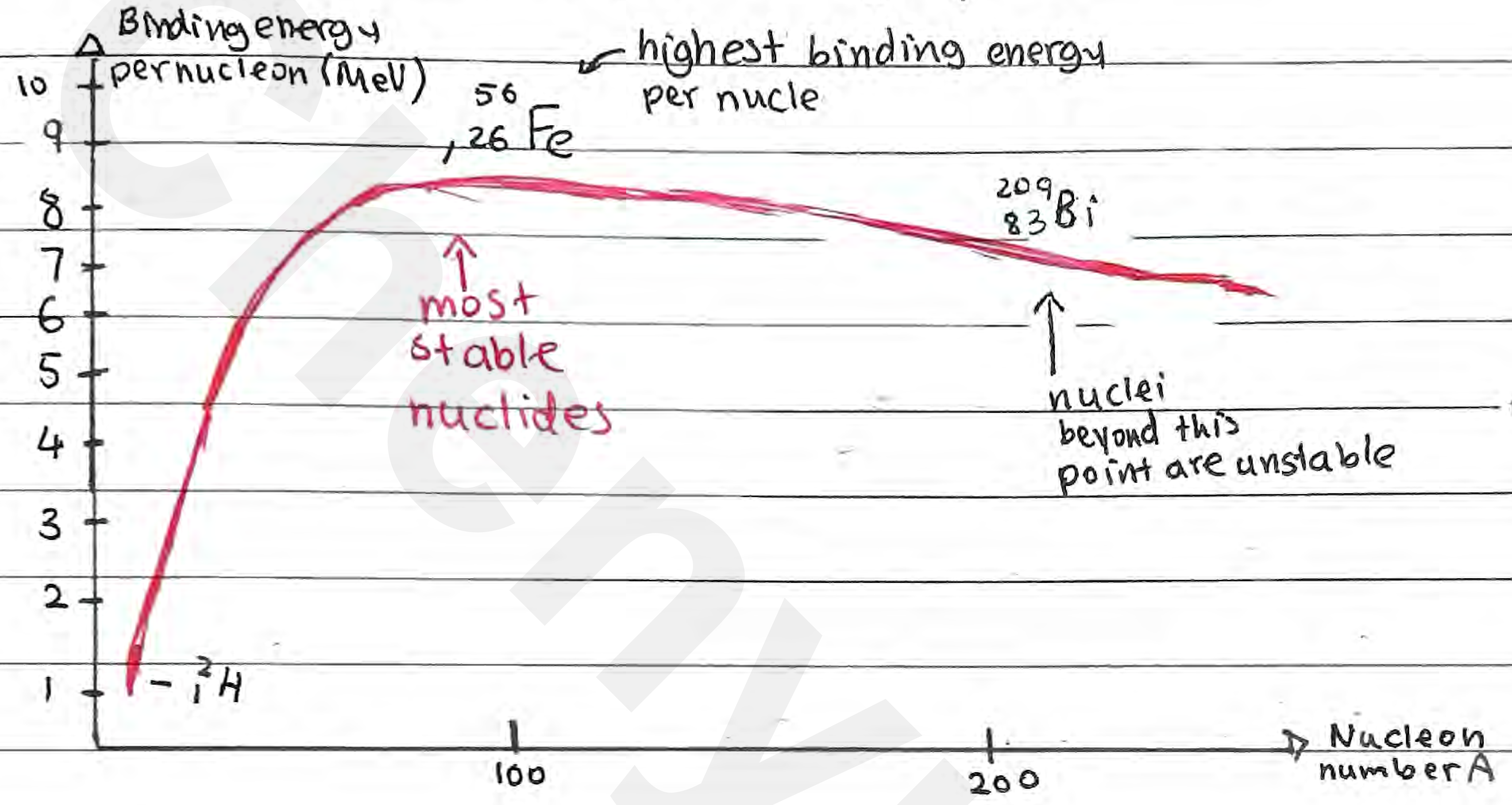
if asking for MeV for energy then... $\frac{BE}{c^2} = \Delta m$ ← use conversion from u to $\frac{MeV}{c^2}$

↳ typically not using Joules in nuclear physics

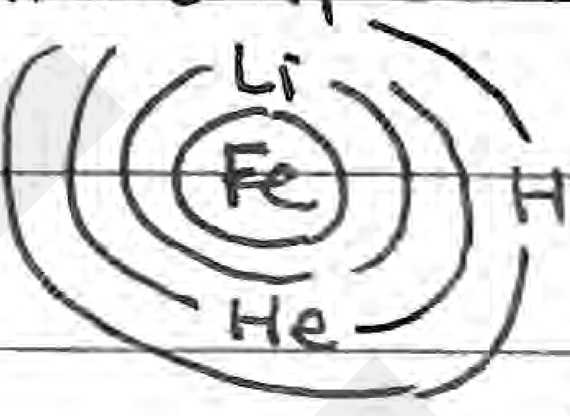
thus... BE = ___ MeV

Binding Energy

When you calculate the binding energy per nucleon for successive elements, you can graph it...



Since $^{56}_{26}Fe$ is the one requiring most binding energy per nucleon, we know it'll be formed in the center of stars:



Stability

- Nuclear stability depends on # of protons with neutrons
- This stability deviates from 1:1 ratio
  (requires more neutrons per proton as # of protons increase)
  ↳ Since greater repulsive electromagnetic force (act on range = ∞)
  ↳ An attractive force is required to overcome it
  - Strong Nuclear Force (range = nucleus ≈ 10^{-10} m; only acts on nucleons)

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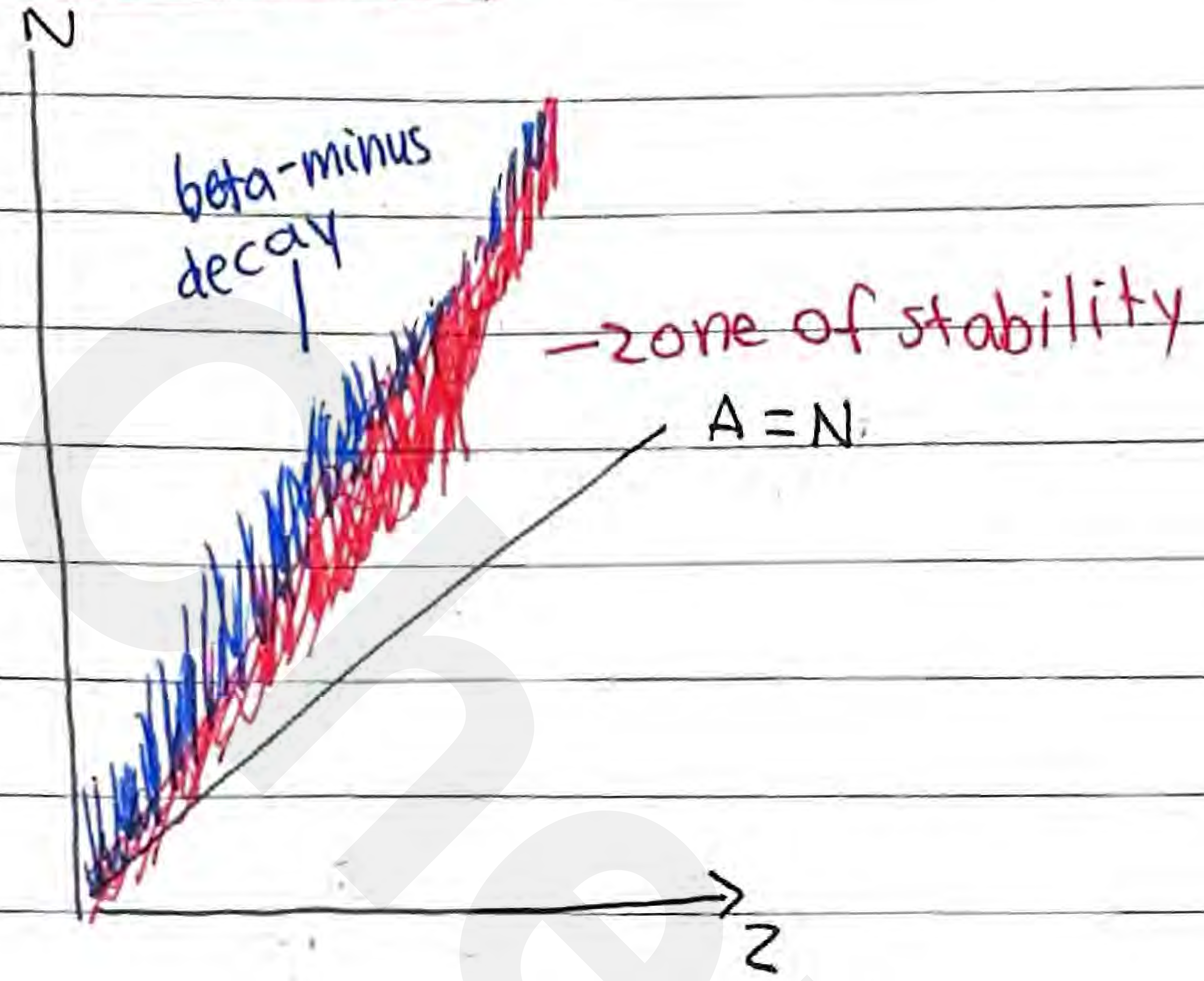
Nuclear Stability



Too little neutrons
↳ Unstable since low Strong Nuclear Force

Too many neutrons
↳ Unstable since Strong Nuclear Force out of range

Stability depends on potential energy
- Low potential energy = stable (ground state)
- High potential energy = unstable (excited state)